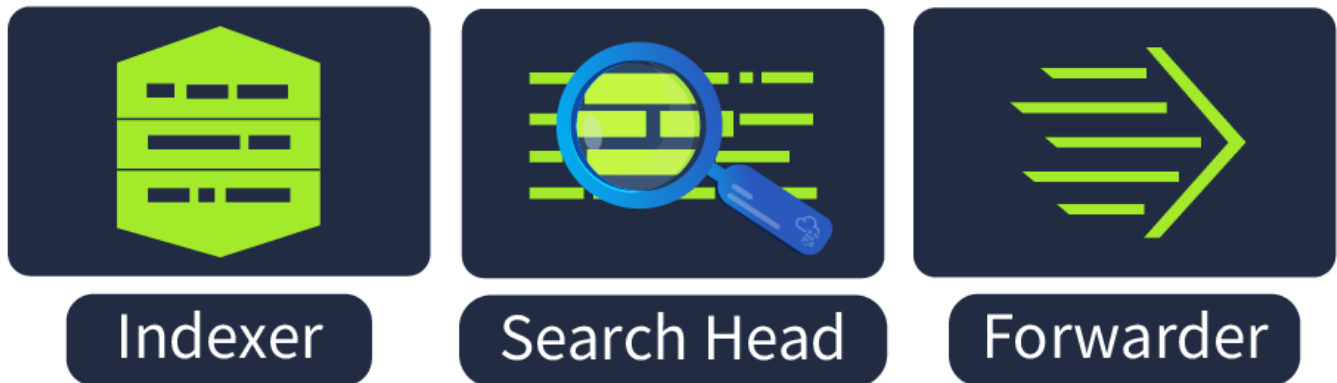


TryHackMe - Splunk Basics

Splunk Components

Splunk has three main components, namely Forwarder, Indexer, and Search Head. These components are explained below:



Splunk Forwarder

Splunk Forwarder is a lightweight agent installed on the endpoint intended to be monitored, and its main task is to collect the data and send it to the Splunk instance. It does not affect the endpoint's performance as it takes very few resources to process. Some of the key data sources are:

- Web server generating web traffic.
- Windows machine generating Windows Event Logs, PowerShell, and Sysmon data.
- Linux host generating host-centric logs.
- Database generating DB connection requests, responses, and errors.

Splunk Indexer

Splunk Indexer plays the main role in processing the data it receives from forwarders. It takes the data, normalizes it into field-value pairs, determines the datatype of the data, and stores them as events. Processed data is easy to search and analyze.

Search Head

Splunk Search Head is the place within the Search & Reporting App where users can search the indexed logs as shown below. When the user searches for a term or uses a Search language known as Splunk Search Processing Language, the request is sent to the indexer and the relevant events are returned in the form of field-value pairs. Search Head also provides the ability to transform the results into presentable tables, visualizations like pie-chart, bar-chart and column-chart.

Adding Data

Splunk can ingest any data. As per the Splunk documentation, when data is added to Splunk, the data is processed and transformed into a series of individual events.

The data sources can be event logs, website logs, firewall logs, etc. Data sources are grouped into categories. Below is a chart listing from the Splunk documentation detailing each data source category.

Data source	Description
Files and directories	Most data that you might be interested in comes directly from files and directories.
Network events	The Splunk software can index remote data from any network port and SNMP events from remote devices.
IT Operations	Data from IT Ops, such as Nagios, NetApp, and Cisco.
Cloud services	Data from Cloud services, such as AWS and Kinesis.
Database services	Data from databases such as Oracle, MySQL, and Microsoft SQL Server.
Security services	Data from security services such as McAfee, Microsoft Active Directory, and Symantec Endpoint Protection.
Virtualization services	Data from virtualization services such as VMWare and XenApp.
Application servers	Data from application servers such as JMX & JMS, WebLogic, and WebSphere.
Windows sources	The Windows version of Splunk software accepts a wide range of Windows-specific inputs, including Windows Event Log, Windows Registry, WMI, Active Directory, and Performance monitoring.
Other sources	Other input sources are supported, such as FIFO queues and scripted inputs for getting data from APIs, and other remote data interfaces.